Technical Report

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A. Approximate Functions for SPD-A and SDD-A in Section VIII of Our Paper

For SPD-A in Example 2 (Example 3), the approximate objective and constraint functions are chosen as:

$$F_0(y;y^{(k)}) = f_0(y) \tag{1}$$

$$F_{i,\mathbf{x}}(\mathbf{x}_i;\mathbf{x}_i^{(k)},y^{(k)}) = f_i(x_{i,1}^{(k)},x_{i,2},y^{(k)}) + \nabla_{x_{i,1}} f_i(\mathbf{x}_i^{(k)},y^{(k)})(x_{i,1}-x_{i,1}^{(k)}) + \frac{\tau_{\mathbf{x}}}{2}(x_{i,1}-x_{i,1}^{(k)})^2, \tag{2}$$

$$F_{i,y}(y;\mathbf{x}_i^{(k)},y^{(k)}) = \frac{\tau_y}{2}(y-y^{(k)})^2 + \nabla_y f_i(\mathbf{x}_i^{(k)},y^{(k)})(y-y^{(k)}), \tag{3}$$

$$G_i(\mathbf{x}_i,y;\mathbf{x}_i^{(k)},y^{(k)}) = c_{i,1}x_{i,2} + c_{i,0} + g_i(\mathbf{x}_i^{(k)},y^{(k)}) + \nabla_{x_{i,1}} g_i(\mathbf{x}_i^{(k)},y^{(k)})(x_{i,1}-x_{i,1}^{(k)}) + \nabla_y g_i(\mathbf{x}_i^{(k)},y^{(k)})(y-y^{(k)}), \tag{4}$$

where $i \in \mathcal{I}$ and $\tau_{\mathbf{x}} > 0$.

For SDD-A in Example 5, the approximate objective and constraint functions are chosen as:

$$F_i(x_i;x_i^{(k)}) = 3a_{i,3}(x_i^{(k)})^2(x_i-x_i^{(k)}) + a_{i,1}x_i + \begin{cases} \frac{\tau}{2}(x_i-x_i^{(k)})^2 + 2a_{i,2}x_i^{(k)}(x_i-x_i^{(k)}), & a_{i,2} \le 0 \\ a_{i,2}x_i^2, & a_{i,2} > 0 \end{cases}, \tag{5}$$

$$G_i(x_i;x_i^{(k)}) = 3b_{i,3}(x_i^{(k)})^2(x_i-x_i^{(k)}) + b_{i,3}(x_i^{(k)})^3 + \frac{L}{2}(x_i-x_i^{(k)})^2 + b_{i,1}x_i + b/I + \begin{cases} 2b_{i,2}x_i^{(k)}(x_i-x_i^{(k)}) + b_{i,2}(x_i^{(k)})^2, & b_{i,2} \le 0 \\ b_{i,2}x_i^2, & b_{i,2} > 0 \end{cases}, \tag{6}$$

where $i \in \mathcal{I}$, $\tau > 0$, and $L \ge 0$.

For SDD-A in Example 6, the approximate objective function is chosen as:

$$F_i(x_i;x_i^{(k)}) = \nabla f_i(x_i^{(k)})(x_i-x_i^{(k)}) + \frac{\tau}{2}(x_i-x_i^{(k)})^2, \tag{7}$$

with $\tau > 0$, and the approximate constraint function $G_i(x_i;x_i^{(k)})$ is chosen as (6), as in Example 5.

B. Detailed Numerical Setup in Section IX of Our Paper

Our experimental environment is Ubuntu 24.04.1. For hardware, GPU is Nvidia GeForce RTX 4090, and CPU is AMD EPYC 9654 Processor.

1) Problem Parameters: For all examples, we randomly and independently generate 10 samples of the problem parameters. Specifically, for Examples 1-3, the samples of $a_{i,j,l}$, $j = 1,2,3$, $b_{i,1}$, and $c_{i,j}$, $j = 0,1$ are generated according to i.i.d. $\mathcal{N}(0,1)$, the samples of $c_{i,2}$ are generated according to the half-normal distribution of $\mathcal{N}(0,1)$, the samples of a and $b_{i,2}$ are generated according to i.i.d. $\mathcal{U}(0,5000)$, and the samples of y_0 are generated according to i.i.d. $\mathcal{U}(0,1)$. For Examples 4-6, the samples of $a_{i,j}$ and $b_{i,j}$ are generated according to i.i.d. $\mathcal{N}(0,1)$. For Example 4 and Examples 5-6, the samples of b are generated according to i.i.d. $\mathcal{N}(0,(0.001)^2)$ and $\mathcal{U}(-0.001,0)$, respectively. Note that for Examples 4-6, the original objective values may be negative or near-zero. To ensure a clear and consistent visualization of the performance comparisons in Fig. 3, we have applied specific constant offsets to the results of all considered algorithms within Examples 4-6. Specifically, constant offsets of 41, 40, and 0.06 are added to the objective values for Examples 4, 5, and 6, respectively. Note that each linear shift is applied uniformly across all algorithms with in each specific example, thereby preserving the relative performance gaps.

2) Algorithm Parameters: For PD-A, SPD-A, DD-A, and SDD-A, we choose stepsizes $\gamma^{(k)} = \frac{1}{\alpha+\beta k^{\epsilon}}, k \ge 1$. For PD-A on Example 3 and DD-A on Example 6, we utilize MATLAB's built-in fmincon function to numerically solve the subproblems. For SPD-A and SDD-A's inner algorithms, (i.e., Algorithms 2 and 5), we choose stepsizes $\gamma_{in}^{(t)} = \gamma_{in}^{(t-1)}(1-\beta_{in}\gamma_{in}^{(t-1)}), t \ge 1$ and the termination criteria $|F(\mathbf{x}^{(t)},y^{(t)};\mathbf{x}^{(k)},y^{(k)}) - F(\mathbf{x}^{(t-1)},y^{(t-1)};\mathbf{x}^{(k)},y^{(k)})| \le \sigma|F(\mathbf{x}^{(t-1)},y^{(t-1)};\mathbf{x}^{(k)},y^{(k)})|$ with $t \le T$ and $|q(\tilde{\mu}^{(t)};\mathbf{x}^{(k)}) - q(\tilde{\mu}^{(t-1)};\mathbf{x}^{(k)})| \le \sigma|q(\tilde{\mu}^{(t-1)};\mathbf{x}^{(k)})|$ with $t \le T$, respectively. For SCAPD, we choose the same approximate functions as in SPD-A, introduce coupling slack variables $\mathbf{Z} \triangleq (z_{i,j})_{i\in\mathcal{I},j=1,2} \in \mathbb{R}^{I\times 2}$ and additional constraints $\sum_{j=1,2} z_{i,j} \le 0$, $i \in \mathcal{I}$ for the convex approximate problem per iteration, and update the coupling variables $\mathbf{Z}$ by utilizing MATLAB's built-in quadprog function to numerically solve Quadratic Program (QP). For SQP, we utilize MATLAB's built-in quadprog function to solve approximate QPs, backtracking line search (with the penalty parameter ρ, the shrinkage factor β, and the sufficient decrease parameter σ) to determine the stepsize, and the Broyden-Fletcher-Goldfarb-Shanno (BFGS) method to update the approximate Hessian matrix. Note that for all implementations involving MATLAB's built-in quadprog function, we employ the interior-point-convex algorithm with both the optimality tolerance and step tolerance set to 10^{-5}. For AL, we utilize MATLAB's built-in fmincon function to solve AL problems constructed by eliminating only the coupling constraints by a penalty (with the penalty parameter ρ, increase factor β, and the sufficient decrease parameter σ). The parameters for all algorithms are listed in Table I. Notably, for all examples, we choose only one set of parameters for the proposed algorithms and 9 different parameter configurations for the baselines. Accordingly, the

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TABLE I: Algorithm parameter.

Algorithm	Algorithm parameter
PD-A	Ex. 1: $\tau = 0$, $\gamma^{(0)} = 1$, $\alpha = 1$, $\beta = 5$, $\epsilon = 1$; Ex. 2: $\tau = 0$, $\gamma^{(0)} = 1$, $\alpha = 1$, $\beta = 5$, $\epsilon = 1$; Ex. 3: $\tau = 5$, $\gamma^{(0)} = 1$, $\alpha = 1$, $\beta = 5$, $\epsilon = 1$.
SPD-A	Ex. 2: $\tau_{\mathbf{x}} = 10^8$, $\tau_y = 0$, $\gamma^{(0)} = 1$, $\alpha = 0$, $\beta = 1$, $\epsilon = 0.1$, $\gamma_{in}^{(0)} = 1$, $\beta_{in} = 0.5$, $\sigma = 0.05$, $T = 10$; Ex. 3: $\tau_{\mathbf{x}} = 10^8$, $\tau_y = 0$, $\gamma^{(0)} = 1$, $\alpha = 0$, $\beta = 1$, $\epsilon = 0.1$, $\gamma_{in}^{(0)} = 1$, $\beta_{in} = 0.5$, $\sigma = 0.05$, $T = 10$.
DD-A	Ex. 4: $\tau = 8$, $\gamma^{(0)} = 0.01$, $\alpha = 3$, $\beta = 1$, $\epsilon = 0.9$; Ex. 5: $\tau = 10$, $\gamma^{(0)} = 0.01$, $\alpha = 3$, $\beta = 1$, $\epsilon = 0.9$; Ex. 66: $\tau = 10$, $\gamma^{(0)} = 1$, $\alpha = 3$, $\beta = 1$, $\epsilon = 0.9$.
SDD-A	Ex. 5: $\tau = 10^{-5}$, $L = 0.1$, $\gamma^{(0)} = 1$, $\alpha = 1$, $\beta = 1$, $\epsilon = 0.1$, $\gamma_{in}^{(0)} = 1$, $\beta_{in} = 0.5$, $\sigma = 0.05$, $T = 10$; Ex. 6: $\tau = 10^{-1}$, $L = 0.1$, $\gamma^{(0)} = 1$, $\alpha = 1$, $\beta = 1$, $\epsilon = 0.1$, $\gamma_{in}^{(0)} = 0.001$, $\beta_{in} = 0.9$, $\sigma = 0.05$, $T = 10$.
SCAPD	Ex. 2: $\tau_{\mathbf{x}} = 10^8$, $\tau_y = 0$, $\gamma^{(0)} = 1$, $\alpha = 0$, $\beta = 1$, $\epsilon = 0.1$, $\gamma_{in}^{(0)} = 1$, $\beta_{in} = [0.1, 0.2, \cdots, 0.9]$, $\sigma = 0.05$, $T = 10$; Ex. 3: $\tau_{\mathbf{x}} = 10^8$, $\tau_{y=0}$, $\gamma^{(0)} = 1$, $\alpha = 0$, $\beta = 1$, $\epsilon = 0.1$, $\gamma_{in}^{(0)} = 1$, $\beta_{in} = [0.1, 0.2, \cdots, 0.9]$, $\sigma = 0.05$, $T = 10$
SQP	Ex. 1: $\rho = [1, 10, 10^2]$, $\beta = [0.06, 0.08, 0.1]$, $\alpha = 0.1$; Ex. 2: $\rho = [1, 10, 10^2]$, $\beta = [0.06, 0.08, 0.1]$, $\alpha = 0.1$; Ex. 3: $\rho = [1, 10, 10^2]$, $\beta = [0.06, 0.08, 0.1]$, $\alpha = 0.1$ Ex. 4: $\rho = [10^{-3}, 10^{-2}, 10^{-1}]$, $\beta = [0, 1, 0.2, 0.3]$, $\alpha = 0.1$ Ex. 5: $\rho = [10^{-3}, 10^{-2}, 10^{-1}]$, $\beta = [0, 1, 0.2, 0.3]$, $\alpha = 0.1$ Ex. 6: $\rho = [10^{-3}, 10^{-2}, 10^{-1}]$, $\beta = [0, 1, 0.2, 0.3]$, $\alpha = 0.1$
AL	Ex. 1: $\rho = [10^{10}, 10^{12}, 10^{14}]$, $\beta = [2, 4, 6]$, $\alpha = 0.25$ Ex. 2: $\rho = [10^{10}, 10^{12}, 10^{14}]$, $\beta = [2, 4, 6]$, $\alpha = 0.25$ Ex. 3: $\rho = [10^{10}, 10^{12}, 10^{14}]$, $\beta = [2, 4, 6]$, $\alpha = 0.25$ Ex. 4: $\rho = [10^{-1}, 10^{0}, 10^{-1}]$, $\beta = [2, 3, 4]$, $\alpha = 0.25$ Ex. 5: $\rho = [10^{-4}, 10^{-3}, 10^{-2}]$, $\beta = [2, 3, 4]$, $\alpha = 0.25$ Ex. 6: $\rho = [10^{-4}, 10^{-3}, 10^{-2}]$, $\beta = [2, 3, 4]$, $\alpha = 0.25$

results of the baselines in Table III of our paper represent the best outcomes obtained from 9 different parameter settings.

3) Initial Points: For each example, we randomly and independently generate 10 initial points for all applicable algorithms with details given below. For Examples 1-3, we randomly generate 10 initial points of y and $x_{i,1}$, $i \in \mathcal{I}$ according to i.i.d. $\mathcal{U}(0,1)$ and $\mathcal{U}(-1,1)$. For Example 1 and Examples 2-3, we randomly generate 10 initial points of Lagrange multipliers $\tilde{\lambda}_i$ and $\tilde{\mu}_i$, $i \in \mathcal{I}$ associated with the coupling constraints according to i.i.d. $\mathcal{U}(-1,1)$ and $\mathcal{U}(0,1)$, respectively. For Example 4-6, we randomly generate 10 initial points of x_i, $i \in \mathcal{I}$ according to i.i.d. $\mathcal{U}(-0.05, 0.05)$. For Example 4 and Examples 5-6, we randomly generate 10 initial points of the Lagrange multipliers $\tilde{\lambda}$ and $\tilde{\mu}$ associated with the coupling constraints according to i.i.d. $\mathcal{U}(-1,1)$ and $\mathcal{U}(0,1)$, respectively. Notably, the initial points of $(\mathbf{x}, y)$ in Examples 1-3 are feasible, whereas the initial points of $\mathbf{x}$ in Examples 4-6 are not guaranteed to satisfy the coupling constraints.

4) Convergence Criteria for Table III: For Examples 1-3, the convergence criterion is given by:

$$\frac{1}{I}\sum_{i\in\mathcal{I}} \max\{0, g_i(\mathbf{x}_i^{(k)}, y^{(k)})\} < 10^{-6}, k = 9, 10,$$

$$\max_{i\in\mathcal{I}} \max\{0, g_i(\mathbf{x}_i^{(k)}, y^{(k)})\} < 10^{-5}, k = 9, 10,$$

$$|f(\mathbf{x}^{(10)}, y^{(10)}) - f(\mathbf{x}^{(9)}, y^{(9)})| \leq 0.05|f(\mathbf{x}^{(9)}, y^{(9)})|.$$

For Examples 4-6, the convergence criterion is given by:

$$\max\{0, g(\mathbf{x}^{(k)})\} < 10^{-2}, k = 9, 10,$$

$$|f(\mathbf{x}^{(10)}) - f(\mathbf{x}^{(9)})| \leq 0.05|f(\mathbf{x}^{(9)})|.$$

5) Best Objective Values in Fig. 2 and Fig. 3: For Examples 1-3, the best objective values are chosen among all 10 iterates that satisfy:

$$\frac{1}{I}\sum_{i\in\mathcal{I}} \max\{0, g_i(\mathbf{x}_i^{(k)}, y^{(k)})\} < 10^{-6},$$

$$\max_{i\in\mathcal{I}} |h_i(\mathbf{x}_i^{(k)}, y^{(k)})| < 10^{-5}.$$

For Examples 4-6, the best objective values are chosen among all 10 iterates that satisfy:

$$\max\{0, g(\mathbf{x}^{(k)})\} < 10^{-2}.$$

C. Extensions of Our Paper

1) Equivalent Nonconvex Problems: We provide the definition of equivalent nonconvex problems and describe some general transformations that yield equivalent nonconvex problems.

Definition 1 (Equivalent Nonconvex Problems): Two nonconvex problems are equivalent if from any stationary point of one, a stationary point of the other is readily found, and vice versa.

Definition 1 extends the definition of equivalent problem for convex problems and is more reasonable for nonconvex problems.

Introducing Linear Equality Constraints: Consider the following nonconvex problem:

$$\begin{aligned} \mathcal{P}_1: \ \min_{\mathbf{x}_1,\mathbf{x}_2} \ & f(\mathbf{x}_1, \mathbf{x}_2) \\ s.t. \ & \tilde{\mathbf{g}}(\mathbf{x}_1, \mathbf{x}_2) \preceq \mathbf{0}, \\ & \tilde{\mathbf{h}}(\mathbf{x}_1, \mathbf{x}_2) = \mathbf{0}, \\ & \mathbf{g}_i(\mathbf{x}_i) \preceq \mathbf{0}, \ i = 1, 2, \\ & \mathbf{x}_i \in \mathcal{X}_i \ i = 1, 2, \end{aligned}$$

where $\mathbf{x}_i \in \mathbb{R}^{n_i}$, $f: \mathbb{R}^{n_1+n_2} \to \mathbb{R}$, $\tilde{\mathbf{g}}: \mathbb{R}^{n_1+n_2} \to \mathbb{R}^{\tilde{r}}$, $\tilde{\mathbf{h}}: \mathbb{R}^{n_1+n_2} \to \mathbb{R}^{\tilde{m}}$, and $\mathbf{g}_i: \mathbb{R}^{n_i} \to \mathbb{R}^r$ are continuously differentiable, and $\mathcal{X}_i$ is nonempty, closed, and convex, for $i = 1, 2$. We replace $\mathbf{x}_1$ with new variable $\mathbf{z} \triangleq (\mathbf{x}_1, \mathbf{x}_2) \in \mathbb{R}^{n_1+n_2}$ and introduce new linear equality constraint $\begin{bmatrix}\mathbf{0}_{n_2\times n_1} & \mathbf{I}_{n_2}\end{bmatrix}\mathbf{z} = \mathbf{x}_2$ and form the following nonconvex problem:

$$\begin{aligned} \mathcal{P}_1': \ \min_{\mathbf{z},\mathbf{x}_2} \ & f(\mathbf{z}) \\ s.t. \ & \mathbf{C}\mathbf{z} = \mathbf{x}_2, \\ & \tilde{\mathbf{g}}(\mathbf{z}) \preceq \mathbf{0}, \\ & \tilde{\mathbf{h}}(\mathbf{z}) = \mathbf{0}, \\ & \mathbf{g}_1(\mathbf{A}\mathbf{z}) \preceq \mathbf{0}, \\ & \mathbf{g}_2(\mathbf{x}_2) \preceq \mathbf{0}, \\ & \mathbf{z} \in \mathcal{Z}, \\ & \mathbf{x}_2 \in \mathcal{X}_2, \end{aligned}$$

$$\sum_{i=1}^{2}(\nabla_{\mathbf{x}_i} f(\mathbf{x}_1^\dagger,\mathbf{x}_2^\dagger)+\nabla_{\mathbf{x}_i}\tilde{\mathbf{g}}(\mathbf{x}_1^\dagger,\mathbf{x}_2^\dagger)\tilde{\boldsymbol{\mu}}+\nabla_{\mathbf{x}_i}\tilde{\mathbf{h}}(\mathbf{x}_1^\dagger,\mathbf{x}_2^\dagger)\tilde{\boldsymbol{\lambda}}+\nabla\mathbf{g}_i(\mathbf{x}_i^\dagger)\boldsymbol{\mu}_i)^T(\mathbf{x}_i-\mathbf{x}_i^\dagger)\geq\mathbf{0},\ \forall(\mathbf{x}_1,\mathbf{x}_2)\in\mathcal{X}_1\times\mathcal{X}_2, \tag{8}$$

$$\tilde{\boldsymbol{\mu}}\odot\tilde{\mathbf{g}}(\mathbf{x}_1^\dagger,\mathbf{x}_2^\dagger)=\mathbf{0},\ \tilde{\mathbf{g}}(\mathbf{x}_1^\dagger,\mathbf{x}_2^\dagger)\preceq\mathbf{0},\ \tilde{\mathbf{h}}(\mathbf{x}_1^\dagger,\mathbf{x}_2^\dagger)=\mathbf{0},\ \boldsymbol{\mu}_i\odot\mathbf{g}_i(\mathbf{x}_i^\dagger)=\mathbf{0},\ \mathbf{g}_i(\mathbf{x}_i^\dagger)\preceq\mathbf{0},\ \mathbf{x}_i^\dagger\in\mathcal{X}_i, i=1,2. \tag{9}$$

$$\begin{aligned}&(\nabla f(\mathbf{z}^\dagger)+\nabla\tilde{\mathbf{g}}(\mathbf{z}^\dagger)\tilde{\boldsymbol{\mu}}+\nabla\tilde{\mathbf{h}}(\mathbf{z}^\dagger)\tilde{\boldsymbol{\lambda}}+\mathbf{A}^T\nabla\mathbf{g}_1(\mathbf{A}\mathbf{z}^\dagger)\boldsymbol{\mu}_1+\mathbf{C}^T\boldsymbol{\lambda})^T(\mathbf{z}-\mathbf{z}^\dagger)\\&+(-\mathbf{I}_{n_2}\boldsymbol{\lambda}+\nabla\mathbf{g}_2(\mathbf{x}_2)\boldsymbol{\mu}_2)^T(\mathbf{x}_2-\mathbf{x}_2^\dagger)\geq\mathbf{0},\ \forall\mathbf{z}\in\mathcal{Z},\ \mathbf{x}_2\in\mathcal{X}_2,\end{aligned} \tag{10}$$

$$\begin{aligned}&\tilde{\boldsymbol{\mu}}\odot\tilde{\mathbf{g}}(\mathbf{z}^\dagger)=\mathbf{0},\ \boldsymbol{\mu}_1\odot\mathbf{g}_1(\mathbf{A}\mathbf{z}^\dagger)=\mathbf{0},\ \boldsymbol{\mu}_2\odot\mathbf{g}_2(\mathbf{x}_2^\dagger)=\mathbf{0},\\&\mathbf{C}\mathbf{z}^\dagger=\mathbf{x}_2^\dagger,\ \tilde{\mathbf{g}}(\mathbf{z}^\dagger)\preceq\mathbf{0},\ \tilde{\mathbf{h}}(\mathbf{z}^\dagger)=\mathbf{0},\ \mathbf{g}_1(\mathbf{A}\mathbf{z}^\dagger)\preceq\mathbf{0},\ \mathbf{g}_2(\mathbf{x}_2^\dagger)\preceq\mathbf{0},\ \mathbf{z}^\dagger\in\mathcal{Z},\mathbf{x}_2^\dagger\in\mathcal{X}_2.\end{aligned} \tag{11}$$

$$\sum_{i\in\mathcal{I}}\left(\nabla_{\mathbf{x}_i}f(\mathbf{x}^\dagger)+\nabla\mathbf{g}_i(\mathbf{x}_i^\dagger)\boldsymbol{\mu}+\nabla\mathbf{h}_i(\mathbf{x}_i^\dagger)\boldsymbol{\lambda}\right)^T(\mathbf{x}_i-\mathbf{x}_i^\dagger),\ \forall\mathbf{x}\in\mathcal{X}, \tag{12}$$

$$\sum_{i\in\mathcal{I}}\boldsymbol{\mu}\odot\mathbf{g}_i(\mathbf{x}_i^\dagger)=\mathbf{0},\ \sum_{i\in\mathcal{I}}\mathbf{g}_i(\mathbf{x}_i^\dagger)\preceq\mathbf{0},\ \sum_{i\in\mathcal{I}}\mathbf{h}_i(\mathbf{x}_i^\dagger)=\mathbf{0},\ \mathbf{x}^\dagger\in\mathcal{X}. \tag{13}$$

$$\begin{aligned}&\sum_{i\in\mathcal{I}}(\nabla_{\mathbf{x}_i}f(\mathbf{x}^\dagger)+\nabla\mathbf{g}_i(\mathbf{x}_i^\dagger)\boldsymbol{\mu}_i+\nabla\mathbf{h}_i(\mathbf{x}_i^\dagger)\boldsymbol{\lambda}_i)^T(\mathbf{x}_i-\mathbf{x}_i^\dagger)\\&+\sum_{i\in\mathcal{I}}(-\boldsymbol{\mu}_i+\boldsymbol{\mu})^T(\mathbf{z}_i-\mathbf{z}_i^\dagger)+(-\boldsymbol{\lambda}_i+\boldsymbol{\lambda})^T(\mathbf{z}_{i+I}-\mathbf{z}_{i+I}^\dagger)\geq\mathbf{0},\ \forall\mathbf{x}\in\mathcal{X},\mathbf{z}\in\mathbb{R}^{rI+mI},\end{aligned} \tag{14}$$

$$\sum_{i\in\mathcal{I}}\boldsymbol{\mu}\odot\mathbf{z}_i^\dagger=\mathbf{0},\ \sum_{i\in\mathcal{I}}\mathbf{z}_i^\dagger\preceq\mathbf{0},\ \sum_{i\in\mathcal{I}}\mathbf{z}_{i+I}^\dagger=\mathbf{0},\ \mathbf{x}^\dagger\in\mathcal{X},\ \boldsymbol{\mu}_i\odot(\mathbf{g}_i(\mathbf{x}_i^\dagger)-\mathbf{z}_i^\dagger)=\mathbf{0},\ \mathbf{g}_i(\mathbf{x}_i^\dagger)-\mathbf{z}_i^\dagger\preceq\mathbf{0},\ \mathbf{h}_i(\mathbf{x}_i^\dagger)-\mathbf{z}_i^\dagger=\mathbf{0},\ i\in\mathcal{I}. \tag{15}$$

where $\mathbf{A}\triangleq\begin{bmatrix}\mathbf{I}_{n_1} & \mathbf{0}_{n_1\times n_2}\end{bmatrix}$, $\mathbf{C}\triangleq\begin{bmatrix}\mathbf{0}_{n_2\times n_1} & \mathbf{I}_{n_2}\end{bmatrix}$, and $\mathcal{Z}\triangleq\mathcal{X}_1\times\mathcal{X}_2$.

Lemma 1: Problem $\mathcal{P}_1$ and Problem $\mathcal{P}_1'$ are equivalent.

Proof: If $(\mathbf{x}_1^\dagger,\mathbf{x}_2^\dagger)$ is a stationary point of Problem $\mathcal{P}_1$, then there exist Lagrange multipliers $\tilde{\boldsymbol{\mu}}\succeq\mathbf{0}$, $\tilde{\boldsymbol{\lambda}}$ and $\boldsymbol{\mu}_i\succeq\mathbf{0}$, $i=1,2$ that together with $(\mathbf{x}_1^\dagger,\mathbf{x}_2^\dagger)$ satisfy (8) and (9), as shown at the top of next page. If $(\mathbf{z}^\dagger,\mathbf{x}_2^\dagger)$ is a stationary point of Problem $\mathcal{P}_1'$, then there exist Lagrange multipliers $\boldsymbol{\lambda}$, $\tilde{\boldsymbol{\mu}}\succeq\mathbf{0}$, $\tilde{\boldsymbol{\lambda}}$, $\boldsymbol{\mu}_1\succeq\mathbf{0}$, and $\boldsymbol{\mu}_2\succeq\mathbf{0}$ that together with $(\mathbf{z}^\dagger,\mathbf{x}_2^\dagger)$ satisfy (10) and (11), as shown at the top of next page. First, suppose that $(\mathbf{x}_1^\dagger,\mathbf{x}_2^\dagger)$ is a stationary point of Problem $\mathcal{P}_1$, and $\tilde{\boldsymbol{\mu}}\succeq\mathbf{0}$, $\tilde{\boldsymbol{\lambda}}$ and $\boldsymbol{\mu}_i\succeq\mathbf{0}$, $i=1,2$ are the corresponding Lagrange multipliers. Let $\mathbf{z}^\dagger=(\mathbf{x}_1^\dagger,\mathbf{x}_2^\dagger)$ and $\boldsymbol{\lambda}=\mathbf{0}$. From (8) and (9), we can show that (10) and (11) hold true for $(\mathbf{z}^\dagger,\mathbf{x}_2^\dagger)$, $\tilde{\boldsymbol{\mu}}$, $\tilde{\boldsymbol{\lambda}}$, $\boldsymbol{\mu}_i$, $i=1,2$, and $\boldsymbol{\lambda}$, which implies that $(\mathbf{z}^\dagger,\mathbf{x}_2^\dagger)$ is a stationary point of Problem $\mathcal{P}_1'$. Next, suppose that $(\mathbf{z}^\dagger,\mathbf{x}_2^\dagger)$ is a stationary point of Problem $\mathcal{P}_1'$, and $\boldsymbol{\lambda}$, $\tilde{\boldsymbol{\mu}}\succeq\mathbf{0}$, $\tilde{\boldsymbol{\lambda}}$, $\boldsymbol{\mu}_1\succeq\mathbf{0}$, and $\boldsymbol{\mu}_2\succeq\mathbf{0}$ are the corresponding Lagrange multipliers. From the definition of $\mathbf{C}$, we can get $(\mathbf{C}^T\boldsymbol{\lambda})^T(\mathbf{z}-\mathbf{z}^\dagger)+(-\mathbf{I}_{n_2}\boldsymbol{\lambda})^T(\mathbf{x}_2-\mathbf{x}_2^\dagger)=\mathbf{0}$ for all $\mathbf{C}\mathbf{z}=\mathbf{x}_2$ and $\boldsymbol{\lambda}$. Thus, (10) can reduce to (8), as shown at the top of the last page, for all $\mathbf{C}\mathbf{z}=\mathbf{x}_2$ and $\boldsymbol{\lambda}$. Let $\mathbf{x}_1^\dagger=\mathbf{A}\mathbf{z}^\dagger$. Then, from (10) and (11), we can show that (8) and (9) hold true for $(\mathbf{x}_1^\dagger,\mathbf{x}_2^\dagger)$, $\tilde{\boldsymbol{\mu}}$, $\tilde{\boldsymbol{\lambda}}$, and $\boldsymbol{\mu}_i\succeq\mathbf{0}$, $i=1,2$, which implies that $(\mathbf{x}_1^\dagger,\mathbf{x}_2^\dagger)$ is a stationary point of Problem $\mathcal{P}_1$. ■

Introducing separable Constraints: We consider the following nonconvex problem:

$$\begin{aligned}\mathcal{P}_2:\ \min_{\mathbf{x}}\ & f(\mathbf{x})\\ s.t.\ & \sum_{i\in\mathcal{I}}\mathbf{g}_i(\mathbf{x}_i)\preceq\mathbf{0},\\ & \sum_{i\in\mathcal{I}}\mathbf{h}_i(\mathbf{x}_i)=\mathbf{0},\\ & \mathbf{x}\in\mathcal{X},\end{aligned}$$

where $\mathbf{x}\triangleq(\mathbf{x}_1,\cdots,\mathbf{x}_I)\in\mathbb{R}^n$ with $\mathbf{x}_i\in\mathbb{R}^{n_i}$, $f:\mathbb{R}^n\to\mathbb{R}$, $\mathbf{g}_i:\mathbb{R}^{n_i}\to\mathbb{R}^r$, and $\mathbf{h}_i:\mathbb{R}^{n_i}\to\mathbb{R}^m$, $i\in\mathcal{I}$ are continuously differentiable, and $\mathcal{X}$ is nonempty, closed, and convex. We introduce new variables $\mathbf{z}_i\in\mathbb{R}^r$ and $\mathbf{z}_{i+I}\in\mathbb{R}^m$, $i\in\mathcal{I}$, new inequality constraints $\mathbf{g}_i(\mathbf{x}_i)-\mathbf{z}_i\preceq\mathbf{0}$, $i\in\mathcal{I}$, and new equality constraints $\mathbf{h}_i(\mathbf{x}_i)-\mathbf{z}_{i+I}=\mathbf{0}$, $i\in\mathcal{I}$, and form the following nonconvex problem:

$$\begin{aligned}\mathcal{P}_2':\ \min_{\mathbf{x},\mathbf{z}}\ & f(\mathbf{x})\\ s.t.\ & \mathbf{g}_i(\mathbf{x}_i)-\mathbf{z}_i\preceq\mathbf{0},\ i\in\mathcal{I}\\ & \sum_{i\in\mathcal{I}}\mathbf{z}_i\preceq\mathbf{0},\\ & \mathbf{h}_i(\mathbf{x}_i)-\mathbf{z}_{i+I}=\mathbf{0},\ i\in\mathcal{I}\\ & \sum_{i\in\mathcal{I}}\mathbf{z}_{i+I}=\mathbf{0},\\ & \mathbf{x}\in\mathcal{X}.\end{aligned}$$

Lemma 2: Problem $\mathcal{P}_2$ and Problem $\mathcal{P}_2'$ are equivalent.

Proof: If $\mathbf{x}^\dagger$ is a stationary point of Problem $\mathcal{P}_2$, then there exist Lagrange multipliers $\boldsymbol{\mu}\succeq\mathbf{0}$ and $\boldsymbol{\lambda}$ that together with $\mathbf{x}^\dagger$ satisfy (12) and (13), as shown at the top of this page. If $(\mathbf{x}^\dagger,\mathbf{z}^\dagger)$ is a stationary point of Problem $\mathcal{P}_2'$, then there exist Lagrange multipliers $\boldsymbol{\mu}_i\succeq\mathbf{0}$, $\boldsymbol{\lambda}_i$, $i\in\mathcal{I}$, $\boldsymbol{\mu}\succeq\mathbf{0}$, and $\boldsymbol{\lambda}$ that together with $(\mathbf{x}^\dagger,\mathbf{z}^\dagger)$ satisfy (14) and (15), as shown at the top of this page. First, suppose that $\mathbf{x}^\dagger$ is a stationary point

of Problem $\mathcal{P}_2$, and $\boldsymbol{\mu}$ and $\boldsymbol{\lambda}$ are the corresponding Lagrange multipliers. Let $\mathbf{z}_i^\dagger = \mathbf{g}_i(\mathbf{x}_i^\dagger)$, $\mathbf{z}_{i+I}^\dagger = \mathbf{h}_i(\mathbf{x}_i^\dagger)$, $\boldsymbol{\mu}_i = \boldsymbol{\mu}$, and $\boldsymbol{\lambda}_i = \boldsymbol{\lambda}$, $i \in \mathcal{I}$. Then, from (12) and (13), we can conclude that $(\mathbf{x}^\dagger, \mathbf{z}^\dagger)$ together with $\boldsymbol{\mu}$, $\boldsymbol{\lambda}$, $\boldsymbol{\mu}_i$, and $\boldsymbol{\lambda}_i$, $i \in \mathcal{I}$, satisfies (14) and (15), which implies that $(\mathbf{x}^\dagger, \mathbf{z}^\dagger)$ is a stationary point of Problem $\mathcal{P}_2'$. Next, suppose that $(\mathbf{x}^\dagger, \mathbf{z}^\dagger)$ is a stationary point of Problem $\mathcal{P}_2'$, and $\boldsymbol{\mu}$, $\boldsymbol{\lambda}$, $\boldsymbol{\mu}_i$, and $\boldsymbol{\lambda}_i$, $i \in \mathcal{I}$ are the corresponding Lagrange multipliers. Substituting $\mathbf{x} = \mathbf{x}^\dagger$ into (14), we have $\sum_{i\in\mathcal{I}}(-\boldsymbol{\mu}_i' + \boldsymbol{\mu})^T(\mathbf{z}_i - \mathbf{z}_i^\dagger) + \sum_{i\in\mathcal{I}}(-\boldsymbol{\lambda}_i' + \boldsymbol{\lambda})^T(\mathbf{z}_{i+I} - \mathbf{z}_{i+I}^\dagger) \geq \mathbf{0}$ for all $\mathbf{z} \in \mathbb{R}^{rI+mI}$, which implies $\boldsymbol{\mu}_i = \boldsymbol{\mu}$ and $\boldsymbol{\lambda}_i = \boldsymbol{\lambda}$, $i \in \mathcal{I}$. Thus, (14) reduces to (12). Then, by eliminating variables $\mathbf{z}_i^\dagger$, $i \in \mathcal{I}$ in (15) and replacing variables $\mathbf{z}_{i+I}^\dagger$ in (15) with $\mathbf{h}_i(\mathbf{x}_i^\dagger)$ based on the equalities $\mathbf{h}_i(\mathbf{x}_i^\dagger) - \mathbf{z}_{i+I}^\dagger = \mathbf{0}$, $i \in \mathcal{I}$, (15) reduces to (13). Thus, (12) and (13) hold true for $\mathbf{x}^\dagger$, $\boldsymbol{\mu}$, and $\boldsymbol{\lambda}$, which implies that $\mathbf{x}^\dagger$ is a stationary point of Problem $\mathcal{P}_2$. ■

2) Indirect Decomposition: Like the convex case, Problem $\mathcal{P}^{pd}$ can be converted to a special case of Problem $\mathcal{P}^{dd}$, and vice versa. Thus, primal and dual decomposition can be indirectly applied to Problem $\mathcal{P}^{dd}$ and Problem $\mathcal{P}^{pd}$, respectively.

Problem $\mathcal{P}^{pd}$, suited for primal decomposition, can be equivalently converted to Problem $\mathcal{P}^{pd-dd}$ by replacing the optimization variables and introducing linear equality constraints:

$$\mathcal{P}^{pd-dd}: \min_{\mathbf{z}} \sum_{i\in\{0\}\cup\mathcal{I}} f_i(\mathbf{z}_i)$$

$$s.t.\ \mathbf{C}_i\mathbf{z}_i = \mathbf{z}_0,\ i \in \mathcal{I}, \tag{16}$$

$$\tilde{\mathbf{h}}_i(\mathbf{z}_i) = \mathbf{0},\ i \in \mathcal{I}, \tag{17}$$

$$\tilde{\mathbf{g}}_i(\mathbf{z}_i) \preceq \mathbf{0},\ i \in \mathcal{I}, \tag{18}$$

$$\mathbf{g}_i(\mathbf{C}_i'\mathbf{z}_i) \preceq \mathbf{0},\ i \in \{0\}\cup\mathcal{I}, \tag{19}$$

$$\mathbf{z}_i \in \mathcal{Z}_i,\ i \in \{0\}\cup\mathcal{I}, \tag{20}$$

where $\mathbf{z}_0 \triangleq \mathbf{y} \in \mathbb{R}^{n_0}$, $\mathbf{C}_0' \triangleq \mathbf{I}_{n_0}$, $\mathcal{Z}_0 \triangleq \mathcal{Y}$, $\mathbf{z}_i \triangleq (\mathbf{x}_i, \mathbf{y}) \in \mathbb{R}^{n_i+n_0}$, $\mathbf{C}_i \triangleq \begin{bmatrix}\mathbf{0}_{n_0\times n_i} & \mathbf{I}_{n_0}\end{bmatrix}$, $\mathbf{C}_i' \triangleq \begin{bmatrix}\mathbf{I}_{n_i} & \mathbf{0}_{n_i\times n_0}\end{bmatrix}$, $\mathcal{Z}_i \triangleq \mathcal{X}_i \times \mathcal{Y}$, $i \in \mathcal{I}$, and the constraints in (16) are additionally introduced. It follows immediately from Lemma 1 that Problem $\mathcal{P}^{pd}$ is equivalent to Problem $\mathcal{P}^{pd-dd}$. Letting $\mathbf{A}_0 \triangleq -\begin{bmatrix}\mathbf{I}_{n_0} & \cdots & \mathbf{I}_{n_0}\end{bmatrix}^T \in \mathbb{R}^{n_0 r\times n_0}$ and $\mathbf{A}_i \triangleq \begin{bmatrix}\mathbf{0} & \cdots & \mathbf{C}_i^T & \cdots & \mathbf{0}\end{bmatrix}^T \in \mathbb{R}^{n_0 r\times(n_i+n_0)}$, $i \in \mathcal{I}$, the constraints in (17) can be rewritten as $\sum_{i\in\{0\}\cup\mathcal{I}} \mathbf{A}_i\mathbf{z}_i = \mathbf{0}$. Let $\tilde{m} \triangleq \sum_{i\in\mathcal{I}} \tilde{m}_i$. Denote $\tilde{\mathbf{H}}_0(\mathbf{z}_0) \triangleq \mathbf{0}$. For all $i \in \mathcal{I}$, denote $\tilde{\mathbf{H}}_i : \mathcal{Z}_i \to \mathbb{R}^{\tilde{m}}$ as

$$\langle\tilde{\mathbf{H}}_i(\mathbf{z}_i)\rangle_j \triangleq \begin{cases}\langle\tilde{\mathbf{h}}_i(\mathbf{z}_i)\rangle_j, & j = 1 + \sum_{i'=1}^{i-1}\tilde{m}_{i'}, \cdots, 1 + \sum_{i'=1}^{i}\tilde{m}_{i'} \\ 0, & \text{otherwise}\end{cases},$$

for $i \in \mathcal{I}$. Then, (17) can be rewritten as $\sum_{i\in\{0\}\cup\mathcal{I}} \tilde{\mathbf{H}}_i(\mathbf{z}_i) = \mathbf{0}$. Thus, Problem $\mathcal{P}^{pd-dd}$ can be viewed as a special case of Problem $\mathcal{P}^{dd}$ without the coupling inequality constraint in (33) of our paper. Consequently, Problem $\mathcal{P}^{pd}$ can be indirectly solved by applying dual decomposition to Problem $\mathcal{P}^{pd-dd}$.

In addition, Problem $\mathcal{P}^{dd}$, suited for dual decomposition, can be equivalently converted to Problem $\mathcal{P}^{dd-pd}$ by introducing new variables, inequality constraints, and equality constraints:

$$\mathcal{P}^{dd-pd}: \min_{\mathbf{x},\mathbf{z}} \sum_{i\in\mathcal{I}} f_i(\mathbf{x}_i)$$

$$s.t.\ \tilde{\mathbf{g}}_i(\mathbf{x}_i) - \mathbf{z}_i \preceq \mathbf{0},\ i \in \mathcal{I}, \tag{21}$$

$$\tilde{\mathbf{h}}_i(\mathbf{x}_i) - \mathbf{z}_{i+I} = \mathbf{0},\ i \in \mathcal{I}, \tag{22}$$

$$\mathbf{z} \in \mathcal{Z}, \tag{23}$$

(33), (36) in our paper,

where $\mathbf{z}_i \in \mathbb{R}^{\tilde{r}}$, $\mathbf{z}_{i+I} \in \mathbb{R}^{\tilde{m}}$, $i \in \mathcal{I}$, and $\mathcal{Z} \triangleq \{\mathbf{z} \in \mathbb{R}^{\tilde{r}I+\tilde{m}I} | \sum_{i\in\mathcal{I}} \mathbf{z}_i \preceq \mathbf{0}, \sum_{i\in\mathcal{I}} \mathbf{z}_{i+I} = \mathbf{0}\}$. Based on Lemma 2, we can show that Problem $\mathcal{P}^{dd}$ is equivalent to Problem $\mathcal{P}^{dd-pd}$. Obviously, Problem $\mathcal{P}^{dd-pd}$ is a special case of Problem $\mathcal{P}^{spd}$ without the decouple inequality constraints for $mathby$ in (13) of our paper and the coupling variable $\mathbf{z}$ in the objective function. Accordingly, Problem $\mathcal{P}^{dd}$ can be indirectly solved by applying primal decomposition to Problem $\mathcal{P}^{dd-pd}$.

3) Two-Level Decomposition: Similar to the convex counterpart, we can repeatedly apply primal and dual decomposition at different levels to decouple problems with both coupling variables and constraints:

$$\mathcal{P}^{tl}: \min_{\mathbf{x},\mathbf{y}} \sum_{i\in\mathcal{I}} f_i(\mathbf{x}_i, \mathbf{y}) + f_0(\mathbf{y})$$

$$s.t.\ \sum_{i\in\mathcal{I}} \tilde{\mathbf{g}}_i(\mathbf{x}_i, \mathbf{y}) \preceq \mathbf{0}, \tag{24}$$

$$\sum_{i\in\mathcal{I}} \tilde{\mathbf{h}}_i(\mathbf{x}_i, \mathbf{y}) = \mathbf{0}, \tag{25}$$

(11)-(14) in our paper,

where $\mathbf{x}$, $\mathbf{y}$, f_i, and $\mathbf{g}_i$, $i \in \{0\}\cup\mathcal{I}$ are the same as in Problem $\mathcal{P}^{pd}$, $\tilde{\mathbf{g}}_i : \mathcal{U}_i \times \mathcal{V} \to \mathbb{R}^{\tilde{r}}$, and $\tilde{\mathbf{h}}_i : \mathcal{U}_i \times \mathcal{V} \to \mathbb{R}^{\tilde{m}}$, $i \in \mathcal{I}$. In the following, we introduce two two-level decompositions for solving Problem $\mathcal{P}^{tl}$.

(i) At the first level, we apply the dual decomposition to deal with the coupling constraints in (24) and (25). Then, at the second level, we use the (successive) primal decomposition to deal with the coupling variable $\mathbf{y}$. Applying the primal decomposition at the second level leads to a two-level optimization decomposition with a master dual problem, a secondary master primal problem, and the subproblems; applying the successive primal decomposition at the second level leads to a two-level optimization decomposition with a master dual problem, a sequence of secondary master primal problems, and a sequence of subproblems.[1]

(ii) At the first level, we use the primal decomposition to deal with the coupling variable $\mathbf{y}$. Then, at the second level, we apply the dual decomposition to deal with the coupling constraints in (24) and (25). Applying the dual decomposition at the second level results in a two-level optimization decomposition with a master primal problem, a secondary master dual problem, and the subproblems.[2]

[1]At the first level, successive dual decomposition is not applicable due to the existence of the generally nonlinear equality constraint in (25).

[2]Successive primal and dual decompositions are not applicable at the first and second levels, respectively, due to the existence of the generally nonlinear equality constraint in (25).